

A Real-Time Multimodal Student Engagement & Behavioural Analytics System

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1. Abstract

In most contemporary classrooms, an educator stands before dozens of students with no definitive, real-time method to verify cognitive comprehension or emotional state. While some students appear visually attentive, they may be mentally checked out; others face confusion or silent anxiety without actively seeking assistance. This project proposes a real-time, AI-powered classroom intelligence system designed to extend an educator's peripheral awareness by monitoring student engagement and distress. Utilizing a standard webcam and microphone, the architecture orchestrates three parallel computer vision streams: facial landmark head-pose analysis, emotion recognition, and eye-tracking alongside an asynchronous speech processing layer running OpenAI Whisper and a fine-tuned Wav2Vec2 model. These five independent signal channels feed into a custom-designed fusion module that applies a rolling, weighted smoothing window to generate a continuous engagement score and a boolean distress flag. Sustained anomalies trigger a generative AI reasoning layer via the Gemini API, converting raw multi-modal metrics into actionable, natural-language teacher alerts and student nudges. Built entirely on open-source frameworks, the primary academic contribution of this thesis is the specific multi-stream fusion logic and a quantitative ablation study evaluating single-modal baselines against the fully integrated multi-modal pipeline. This system establishes a foundational methodology for real-time affective computing in edge-deployable educational systems and automated student welfare monitoring.

2. Introduction and Motivation

Educational monitoring has historically relied on passive, retrospective assessment paradigms. Instructors traditionally gauge student comprehension via reactive signals, such as hand-raising, or lagging metrics like exam performance and drop-out rates, at which point the optimal window for targeted academic intervention has passed. This operational deficit has intensified with the widespread adoption of synchronous online learning and

hybrid classrooms. Within digital environments, instructors are isolated from organic peripheral cues, reduced to observing thumbnail-sized video feeds where micro-expressions of fatigue or cognitive overload are easily lost. Even within physical classrooms, human attention does not scale efficiently; managing a room of thirty or more distinct learners prevents simultaneous, individual behavioral tracking.

While computer vision frameworks have successfully automated specific isolated metrics like head-pose or basic facial expression recognition, standalone modalities remain fragile and prone to misinterpretation. For instance, a student can maintain rigid head orientation toward a whiteboard while experiencing total cognitive disengagement—a nuance a basic pose estimator misses. Conversely, a facial emotion model might misinterpret natural resting expressions as active resistance or boredom. This project is motivated by the understanding that a student's true internal state is a multi-modal expression: a fluid combination of gaze direction, blink frequency, facial muscular changes, vocal prosody, and verbal semantic choices. Capturing and aligning these synchronous features yields a robust, context-aware proxy of learner distress. Beyond typical classrooms, this framework addresses critical accessibility gaps in special needs education, acting as a non-verbal communication bridge for learners with neurodivergent profiles (e.g., ADHD, autism, or severe social anxiety) who may be fundamentally unable to self-report distress during periods of frustration.

3. Literature Review

Automated engagement detection has evolved from static, rule-based systems to dynamic deep learning pipelines over the past two decades. Early foundational research heavily utilized Ekman and Friesen's Facial Action Coding System (FACS) to classify standard emotional states. Models trained on constrained datasets like CK+ or DISFA achieved high accuracy under laboratory settings but regularly failed in real-world classroom environments due to uneven lighting, user movement, and the subtle nature of cognitive

affect. Active learning rarely elicits exaggerated expressions; instead, it manifests as minor brow furrows, wandering focus, or drooping eyelids.

To track attention direction, researchers introduced head-pose estimation as a primary feature. The release of open-source libraries like Google's MediaPipe Holistic—capable of tracking 468 dense facial landmarks in real time on standard consumer hardware—democratized this approach. Work by Liao et al. (2021) demonstrated that spatial vectors representing yaw, pitch, and roll derived directly from these facial meshes correlate tightly with self-reported student attention spans. Nonetheless, the fundamental limitation of using head pose as an exclusive proxy for focus is its vulnerability to passive compliance, where a student mimics forward-facing posture while suffering from internal distraction. Eye-tracking and gaze estimation have been layered into vision systems to mitigate this error. Blink rate is an established physiological marker for cognitive fatigue and drowsiness, while coordinate-based gaze mapping tracks true focus shifts relative to a screen or presentation space. Historically, these metrics demanded costly, specialized infrared hardware, but recent developments in webcam-driven landmark tracking have bypassed these physical deployment constraints.

Concurrently, audio-centric deep learning has progressed along an isolated vector. Self-supervised models like Wav2Vec2.0, pre-trained on massive speech corpuses and fine-tuned on emotional audio datasets such as RAVDESS, can accurately categorize vocal prosody into discrete emotional states like frustration, calm, or fear. Furthermore, advanced speech-to-text engines like OpenAI's Whisper allow for the extraction of semantic content from student utterances in real time, exposing explicit linguistic markers of confusion.

Prior research by Kaur et al. (2018) on affect prediction in unconstrained environments and Monkaresi et al. (2017) on combining visual expressions with heart rate data demonstrates the utility of multi-channel data collection. However, these systems generally suffer from high computational overhead, require intrusive physiological sensors, or function strictly via offline batch-processing. A major gap remains: existing architectures fail to

concurrently execute light, multi-stream vision and audio pipelines on consumer-tier devices, fuse them through an explicit weighted algorithmic layer, and pipe the resulting state vector through a large language model (LLM) reasoning agent to deliver natural-language insights. This project addresses this specific engineering and analytical integration challenge.

4. Problem Statement and Objectives

Problem Statement

Real-time, multi-modal student engagement and distress monitoring in real-world educational settings remains a difficult engineering and deployment challenge. Existing setups are typically single-modal and fragile, or multi-modal but structurally impractical requiring specialized, expensive hardware or offline processing pipelines that produce complex mathematical outputs which educators cannot readily interpret or apply mid-lesson. There is a distinct lack of lightweight, edge-compatible architectures that integrate vision, gaze, and voice streams into a unified, smoothed engagement score while leveraging generative models to translate abstract telemetry into immediate, plain-language teaching prompts.

Objectives

1. Develop and Optimize Parallel Vision Streams

Implement a multi-threaded visual ingestion engine comprising MediaPipe head-pose tracking, DeepFace/FER expression analysis, and geometric eye-tracking running simultaneously on a commodity webcam feed.

3. Design a Unified Multi-Modal Fusion Engine

Formulate a specialized algorithmic module to map all five asynchronous signal streams into a single engagement metric (ranging from 0 to 100) and a binary

distress flag, utilizing a configurable weighted schema and a rolling ten-second window to filter structural noise.

4. Integrate GenAI Reasoning Layer

Program an automated prompting pipeline connecting the structured fusion outputs to the Gemini API to synthesize context-aware, non-alarming natural-language suggestions for the instructor every thirty seconds.

5. Design a Responsive Real-Time Classroom Dashboard

Build a low-latency, intuitive web user interface that translates complex multi-modal metrics into clean visual heatmaps and dynamic trend charts, ensuring educators can interpret student attention drops and emotional distress flags at a single glance.

5. Methodology and Planning of Work

System Architecture

The technical architecture is split across four distinct programmatic layers operating over a continuous streaming pipeline:

Layer 1 — Vision (Parallel Extraction)

- Every incoming webcam frame is processed in parallel by three visual components. Stream A uses MediaPipe Face Mesh to extract 468 landmark coordinates to compute 3D head rotation (yaw, pitch, roll). Stream B runs a lightweight DeepFace/FER inference to categorize micro-expressions across seven primary emotional states. Stream C processes eye landmark geometric variations to calculate instantaneous blink frequency and gaze drift indexes.

Layer 2 — Multi-Modal Fusion Module

- The programmatic core of the project ingests the raw metrics from the visual and acoustic layers, standardizing them into a single analytical vector. It computes a composite engagement index utilizing an explicit weighting matrix

Layer 3 — LLM Interpretation & Serving

- Every thirty seconds, this moving vector is formatted into a structured JSON payload and sent to the Gemini API along with an instruction prompt. The model interprets trends across the numerical data and generates context-aware, conversational coaching alerts or student prompts.

Project Implementation Plan

Phase	Weeks	Activity Explanation
Project planning	Week 1	Setting up repositories and configuring environment dependencies like MediaPipe, SNN Torch, and Tonic.
Literature Review	Week 1	Analyzing foundational research on FACS, landmark extraction, and spatial transformation matrices for head-pose baselines.
Data Collection, Annotation, Preparing	Weeks 2–3	Capturing multi-stream video landmarks and frame extraction and perform the preparation
Model Training - Testing, Modify	Weeks 4–5	Developing the custom fusion.py weighted script and iteratively engineering prompts for the Gemini API reasoning layer.
Analysing - Deploy	Weeks 6-7	Wrapping the multi-threaded pipeline in FastAPI endpoints and web user interface that translates complex multi-modal metrics into clean visual
Report Drafting	Week 8	Compiling the final 5-chapter research thesis

6. Facilities Required for Proposed Work

Computing Hardware: Standard development workstation equipped with a minimum of 8 GB RAM and an integrated video capture device. The system does not require a discrete GPU, as frame-skipping techniques are used to keep inference light. A standard audio microphone is used for voice tracking.

Language Ecosystem & Core Frameworks: Python 3.10+ containing MediaPipe for facial landmark extraction, DeepFace for emotion detection, OpenAI Whisper for low-latency voice transcription, and HuggingFace Transformers for running local Wav2Vec2 models.

Reference Datasets: The RAVDESS audio dataset is used to calibrate the vocal emotion model, and the FER2013 image dataset serves as an internal benchmark to validate visual emotion accuracy during testing.

7. References

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